Li Liu

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Ph.D. Candidate in Computer Science and Engineering, UC Santa Cruz

Open to full-time roles starting 2027
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Multimodal LLMs · Agents · Spatial reasoning

Research Projects

Spatial Intelligence (COLM 2026, NeurIPS 2024, one paper under review):
Toward AI that perceives space, reasons about it, and augments human spatial perception, by making spatial representations explicit so they can be measured, trained, and checked by people.

- **Perceive Space:** Defined a new task, directional guidance, and built and open-sourced its 821-sample benchmark of real photos taken by blind users: VLMs answer directly even when the image cannot support an answer. Designed a label-free synthetic-data pipeline that works on any VQA dataset; fine-tuning on it raised open VLMs by up to 16 accuracy points, the best beating GPT-4o. [link]
- **Represent Space:** Built a benchmark that separates building a spatial layout from reasoning over it, and tested six LLMs: all read layouts better than they build them. Fine-tuning models to draw layouts more than doubled construction accuracy and improved text-only spatial reasoning, transferring to three external benchmarks; released the dataset. [link]
- **Enhance Human Perception:** Led the project from zero: with a blind co-designer, built a voice-and-touch map agent on Gemini's native-audio model that runs on an everyday laptop, benchmarked its spatial answers (83% correct on 393 spoken questions), ran 2-hour in-person studies with 14 blind and low-vision adults aged 23 to 81, and led the qualitative analysis. Half found places they had not asked about. Demoed at PitchAbility 2026. [link]

Uncertainty-Aware Prediction (IEEE TIM, EJR Open): Proposed CPSC, a conformal predictor more accurate at lower compute than conformal k-NN, whose reliability scores select unlabeled data for augmentation [link]; applied conformal prediction to COVID-19 ICU-risk triage that held up on new variants [link].

Selected Publications

[1] **S. Huang*, L. Liu*** et al. "Learning to Draw ASCII Improves Spatial Reasoning in Language Models." *COLM* 2026. (*co-first author) [link]

[2] **L. Liu** et al. "Right This Way: Can VLMs Guide Us to See More to Answer Questions?" *NeurIPS* 2024. [link]

[3] **L. Liu** et al. "TouchingSpace: An Audio-Haptic Map for Blind and Low-Vision Readers." *Under review; arXiv* 2026. [link]

[4] **L. Liu** et al. "Pattern Discovery with Wide-Lens Analysis and Sharp-Focus Validation." *WWW Demo Track* 2026. [link]

[5] **L. Liu** et al. "CPSC: Conformal Prediction With Shrunken Centroids for Efficient Prediction Reliability Quantification and Data Augmentation." *IEEE TIM* 2022. [link]

Experience

Amazon — Seattle, WA
Applied Scientist Intern — *June 2026 – Sept. 2026*
- **Skills on Board: Maintaining Agent Skills as Typed Data (under review)** [link]: Proposed and built a system that stores agent skills as typed, versioned data instead of free text, so people can trace and edit each step and agents follow the same source. Halved agent output tokens on SkillLearnBench at equal or better task performance.
- **Pipeline-Wide Failure Attribution**: Linked each skill to its evidence and rollouts so a failure can be traced to the agent, the task, or the benchmark; this found defects in SkillLearnBench itself.

Amazon — Santa Clara, CA
Applied Scientist Intern — *June 2025 – Sept. 2025*
- **Taxonomy Discovery from Unlabeled Conversations (WWW 2026 Demo)**: Designed a modular, human-in-the-loop framework that evolves its own taxonomy from a few seed keywords, alternating a wide-lens view of where patterns cluster with sharp-focus checks of each extraction; every pattern traces to its source.

Amazon — Palo Alto, CA
Applied Scientist Intern — *June 2024 – Sept. 2024*
- **Evidence-Grounded Search from Reviews**: Built a search system that turns customer reviews into search facets, each linked to the review sentence behind it, to answer needs catalogs do not record, such as a dish kids will eat. Later shown on recipes to find 18% more matches than keyword search.

Nissan Research Center — Santa Clara, CA
Data Scientist Intern — *July 2023 – Sept. 2023*
- **Driver Engagement in Autonomous Driving**: Developed a time-series model of driver engagement from real-world driving logs, improving prediction accuracy by 16 points.

Education

UC Santa Cruz, Ph.D. in Computer Science and Engineering, advisor: Leilani Gilpin — 2022 – 2027 (est.)

Zhejiang University, M.Eng. in Control Science and Engineering — 2019 – 2022

Zhejiang University, B.S., Chu Kochen Honors College — 2015 – 2019

Academic Activities

Reviewer: NeurIPS, ICLR, CVPR, ECCV, CIKM, WACV. **TA:** Python, Computer Vision. **Mentor:** Foothill SLI, UCSC SIP, Stanford summer research.